Step 4 : Consider the test statistic $Z = \dfrac{(p_X - p_Y) - (P_X - P_Y)}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{m} + \frac{1}{n}\right)}}$ under H_0. Here, $\hat{p} = \dfrac{mp_X + np_Y}{m+n}$, $\hat{q} = 1 - \hat{p}$. The approximate sampling distribution of the test statistic under H_0 is the $N(0,1)$ distribution.

Step 5 : Calculate the value of Z for the given data as $z_0 = \dfrac{p_X - p_Y}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{m} + \frac{1}{n}\right)}}$.

Step 6 : Choose the critical value, z_e, corresponding to α and H_1 from the following table

Alternative Hypothesis (H_1)	$P_X \neq P_Y$	$P_X > P_Y$	$P_X < P_Y$
Critical Value (z_e)	$z_{\alpha/2}$	z_α	$-z_\alpha$

Step 7 : Decide on H_0 choosing the suitable rejection rule from the following table corresponding to H_1.

Alternative Hypothesis (H_1)	$P_X \neq P_Y$	$P_X > P_Y$	$P_X < P_Y$
Rejection Rule	$\lvert z_0 \rvert \geq z_{\alpha/2}$	$z_0 > z_\alpha$	$z_0 < -z_\alpha$

Example 1.13

A study was conducted to investigate the interest of people living in cities towards self-employment. Among randomly selected 500 persons from City-1, 400 persons were found to be self-employed. From City-2, 800 persons were selected randomly and among them 600 persons are self-employed. Do the data indicate that the two cities are significantly different with respect to prevalence of self-employment among the persons? Choose the level of significance as $\alpha = 0.05$.

Solution:

Step1 : Let P_X and P_Y be respectively the proportions of self-employed people in City-1 and City-2. Then, the null and alternative hypotheses are

Null hypothesis: $H_0 : P_X = P_Y$

i.e., there is no significant difference between the proportions of self-employed people in City-1 and City-2.

Alternative hypothesis: $H_1 : P_X \neq P_Y$

i.e., difference between the proportions of self-employed people in City-1 and City-2 is significant. It is a two-sided alternative hypothesis.

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